Mathematical Analysis — Midterm 2024, Row 2

Past exam · Răzvan

> **How to use.** The statement is on this page, exactly as in the official PDF (Class Materials → Past Exams → *midterm_2024_2.pdf*). Solve it on paper in **1h50min**, then turn to the solutions, which come after a red STOP page.

Midterm Test — Row 2

1. Let $A \subseteq \mathbb{R}$. Prove that $\operatorname{int} A \subseteq A \subseteq \operatorname{cl} A$. Give an example with strict inclusions. 1p
2. Find the value of $\lim\limits_{n\to\infty}\left(\dfrac{1}{2n+1}+\dfrac{1}{2n+2}+\cdots+\dfrac{1}{3n}\right)$. 1p
3. Study the convergence of the following: (a) $\displaystyle\sum_{n=2}^{\infty}\frac{1}{n(\ln n)^2}$. (b) $\displaystyle\int_0^{\pi}\frac{1}{\sqrt{\sin x}}\,dx$. (c) $\displaystyle\sum_{n=1}^{\infty}\frac{1\cdot 3\cdot\ldots\cdot(2n-1)}{2^n\,n!}$. 3p
4. Find the Taylor series around 0 and its radius of convergence for each of the following: (a) $\ln(1+x)$ (1p); (b) $1/(1+x^2)$ (1p).
5. Consider the power series $\displaystyle\sum_{n=1}^{\infty}\frac{x^n}{n^2+n}$. (a) Find its radius of convergence and its convergence set. (1p) (b) Find its sum for any x in the convergence set. (1p)

Time 1h50min. Marking starts from 1p. Total 10p.

⛔ STOP
SOLUTIONS BELOW

De aici încep rezolvările

Rezolvă mai întâi subiectul singur, cronometrat (1h50min), și abia apoi dă pagina. Toate paginile cu rezolvări au o bandă roșie în stânga.

SOLUTIONS — Midterm 2024, Row 2

1. Interior and closure

Recall: x is an *interior point* of A if $\exists \varepsilon > 0$ with $(x-\varepsilon, x+\varepsilon) \subseteq A$; x is an *adherent point* (point of closure) of A if every neighbourhood $(x-\varepsilon, x+\varepsilon)$ meets A. $\operatorname{int} A$ and $\operatorname{cl} A$ are the sets of these points.

$\operatorname{int} A \subseteq A$**:** if $x \in \operatorname{int} A$, then $x \in (x-\varepsilon, x+\varepsilon) \subseteq A$, so $x \in A$.

$A \subseteq \operatorname{cl} A$**:** if $x \in A$, then every neighbourhood of x contains x itself, so it meets A; hence $x \in \operatorname{cl} A$. ■

Example: $A = [0,1)$. Then $\operatorname{int} A = (0,1)$ (0 is not interior: every $(-\varepsilon, \varepsilon)$ contains negative numbers) and $\operatorname{cl} A = [0,1]$ (1 is adherent: $(1-\varepsilon, 1+\varepsilon)$ contains points of A).

$\Longrightarrow (0,1) \subsetneq [0,1) \subsetneq [0,1]$**.**

Another standard example: $A = \mathbb{Q}$, with $\operatorname{int} \mathbb{Q} = \emptyset$ and $\operatorname{cl} \mathbb{Q} = \mathbb{R}$.

2. A limit as a Riemann sum

There are n terms:

$$\sum_{k=1}^{n} \frac{1}{2n+k} = \frac{1}{n} \sum_{k=1}^{n} \frac{1}{2+\frac{k}{n}} \longrightarrow \int_0^1 \frac{dx}{2+x} = \ln(2+x)\Big|_0^1 = \ln \frac{3}{2},$$

since this is a Riemann sum of the continuous function $\frac{1}{2+x}$ on $[0,1]$ (points $\frac{k}{n}$, step $\frac{1}{n}$).

$\Longrightarrow$ **the limit is** $\ln \frac{3}{2} \approx 0.4055$**.**

3. Convergence

(a) Positive, decreasing terms $\Longrightarrow$ Cauchy condensation:

$$2^k a_{2^k} = \frac{2^k}{2^k (\ln 2^k)^2} = \frac{1}{k^2 (\ln 2)^2} \quad \Longrightarrow \quad \sum_k \frac{1}{k^2 \ln^2 2} \text{ converges } (p = 2 > 1).$$

$\Longrightarrow \sum \frac{1}{n(\ln n)^2}$ **converges.**

Integral test: $\int_2^\infty \frac{dx}{x \ln^2 x} = -\frac{1}{\ln x}\Big|_2^\infty = \frac{1}{\ln 2} < \infty$. Compare with Row 1: $\frac{1}{n \ln n}$ diverges — the square on $\ln$ makes the difference.

(b) The integral is improper at 0 and at π, and it is positive. By symmetry ($\sin(\pi - x) = \sin x$) it is enough to study $\int_0^{\pi/2}$. On $[0, \frac{\pi}{2}]$ we have $\sin x \geq \frac{2}{\pi} x$ (concavity of $\sin$: the graph lies above the chord from $(0,0)$ to $(\frac{\pi}{2}, 1)$). Hence

$$0 < \frac{1}{\sqrt{\sin x}} \leq \sqrt{\frac{\pi}{2}} \cdot \frac{1}{\sqrt{x}}, \qquad \int_0^{\pi/2} \frac{dx}{\sqrt{x}} = 2\sqrt{x}\Big|_0^{\pi/2} < \infty.$$

By comparison, $\int_0^{\pi/2} \frac{dx}{\sqrt{\sin x}}$ converges, and so does $\int_{\pi/2}^{\pi}$ (substitute $x \mapsto \pi - x$).

$\Longrightarrow$ **the integral converges.**

Equivalently, $\frac{1/\sqrt{\sin x}}{1/\sqrt{x}} = \sqrt{\frac{x}{\sin x}} \to 1$ as $x \to 0^+$, so the integral behaves like $\int_0 \frac{dx}{x^{1/2}}$, which converges ($p = \frac{1}{2} < 1$).

(c) Since $2^n n! = 2 \cdot 4 \cdots (2n)$, $a_n = \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)}$. Ratio: $\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2} \to 1$ (inconclusive). Raabe–Duhamel:

$$n\left(\frac{a_n}{a_{n+1}} - 1\right) = n \cdot \frac{(2n+2)-(2n+1)}{2n+1} = \frac{n}{2n+1} \to \frac{1}{2} < 1.$$

$\Longrightarrow$ **the series diverges.**

4. Taylor series

(a) From $\frac{1}{1+t} = \sum_{n\geq 0}(-1)^n t^n$ ($|t|<1$), integrate on $[0,x]$:

$$\ln(1+x) = \sum_{n\geq 0}(-1)^n \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$\Longrightarrow$ $R = 1$ **(equality on** $(-1,1]$**; at** $x=1$ **it gives** $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \dots$**).**

(b) Geometric series with ratio $-x^2$:

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty}(-x^2)^n = \sum_{n=0}^{\infty}(-1)^n x^{2n}, \qquad \text{valid iff } |x^2| < 1.$$

$\Longrightarrow$ $R = 1$**. (Although** $\frac{1}{1+x^2}$ **is defined on all of** $\mathbb{R}$**, the series diverges for** $|x| \geq 1$**.)**

5. Power series

(a) $c_n = \frac{1}{n(n+1)}$, $\frac{c_n}{c_{n+1}} = \frac{n+2}{n} \to 1 \Longrightarrow R = 1$. At $x = \pm 1$: $\sum \frac{1}{n(n+1)}$ converges, so the series converges absolutely at both endpoints.

$\Longrightarrow$ $R = 1$**, convergence set** $[-1,1]$**.**

(b) $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. For $0 < |x| < 1$:

$$\sum_{n\geq 1}\frac{x^n}{n} = -\ln(1-x), \qquad \sum_{n\geq 1}\frac{x^n}{n+1} = \frac{1}{x}\sum_{m\geq 2}\frac{x^m}{m} = \frac{1}{x}\big(-\ln(1-x) - x\big).$$

$$S(x) = -\ln(1-x) + \frac{\ln(1-x)}{x} + 1 = 1 + \frac{(1-x)\ln(1-x)}{x}.$$

Special points: $S(0) = 0$ (all terms vanish; the formula also tends to 0 as $x \to 0$). $x = -1$: the formula holds by Abel's theorem, $S(-1) = 1 - 2\ln 2$. $x = 1$: the sum telescopes, $S(1) = \sum\left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$.

$\Longrightarrow$ $S(x) = 1 + \frac{(1-x)\ln(1-x)}{x}$ **for** $x \in [-1,1) \setminus \{0\}$**,** $S(0) = 0$**,** $S(1) = 1$**.**

Results at a glance

Ex.	Answer	Tool
1	$A = [0,1)$: $(0,1) \subsetneq [0,1) \subsetneq [0,1]$	definitions
2	$\ln\frac{3}{2}$	Riemann sum
3	(a) converges; (b) converges; (c) diverges	condensation; comparison with $x^{-1/2}$; Raabe ($\frac{1}{2} < 1$)
4	$\sum(-1)^{n-1}\frac{x^n}{n}$; $\sum(-1)^n x^{2n}$; both $R = 1$	geometric series
5	$[-1,1]$; $S = 1 + \frac{(1-x)\ln(1-x)}{x}$	partial fractions